Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY AND MANAGEMENT, PUNE-09

Name : **Class : B.E.** **Roll No :**

PRACTICAL NO. 03

Support Vector Machine (SVM) for Handwritten Digit Classification

Aim

To implement a Support Vector Machine with an RBF kernel that classifies handwritten digits (0–9) from the scikit-learn digits dataset of 8×8 images, and to evaluate it using accuracy, a confusion matrix and a classification report.

Code and Output

In [1]:
```
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (accuracy_score, confusion_matrix,
                             classification_report, ConfusionMatrixDisplay)
```

In [2]:
```
digits=load_digits()
X_train,X_test,y_train,y_test=train_test_split(
    digits.data,digits.target,test_size=.25,random_state=42,
    stratify=digits.target)
model=make_pipeline(StandardScaler(),SVC(kernel="rbf",C=5,gamma="scale"))
model.fit(X_train,y_train)
pred=model.predict(X_test)
print("Accuracy:",round(accuracy_score(y_test,pred),4))
print("Confusion matrix:\n",confusion_matrix(y_test,pred))
print(classification_report(y_test,pred,zero_division=0))
```

```
Accuracy: 0.98
Confusion matrix:
 [[45  0  0  0  0  0  0  0  0  0]
 [ 0 45  0  0  1  0  0  0  0  0]
 [ 0  0 44  0  0  0  0  0  0  0]
 [ 0  0  0 46  0  0  0  0  0  0]
 [ 0  0  0  0 44  0  0  1  0  0]
 [ 0  0  0  0  0 45  0  0  0  1]
 [ 0  0  0  0  0  0 45  0  0  0]
 [ 0  0  0  0  0  1  0 44  0  0]
 [ 0  2  0  0  1  0  0  0 40  0]
 [ 0  0  0  0  0  0  1  1  0 43]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        45
           1       0.96      0.98      0.97        46
           2       1.00      1.00      1.00        44
```

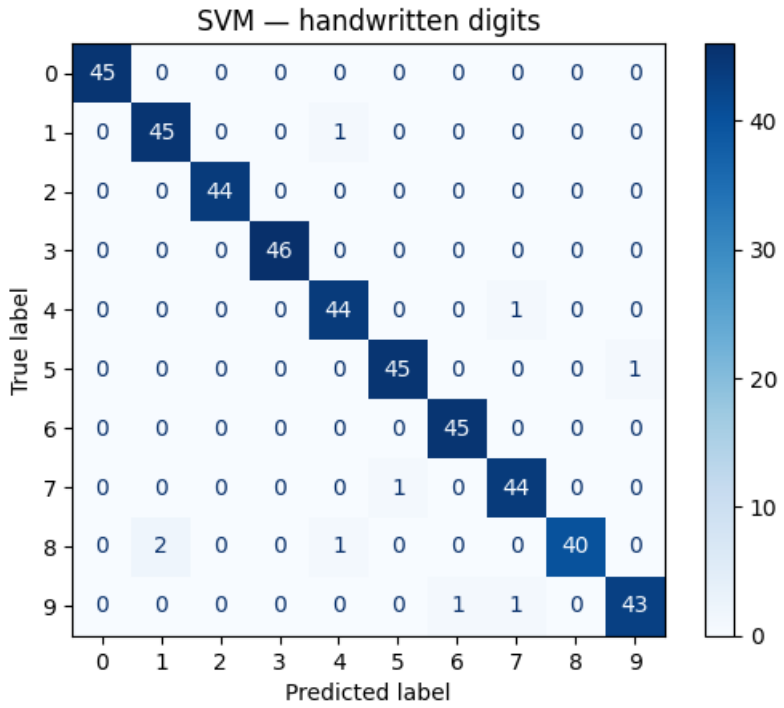
```
              3       1.00      1.00      1.00        46
              4       0.96      0.98      0.97        45
              5       0.98      0.98      0.98        46
              6       0.98      1.00      0.99        45
              7       0.96      0.98      0.97        45
              8       1.00      0.93      0.96        43
              9       0.98      0.96      0.97        45

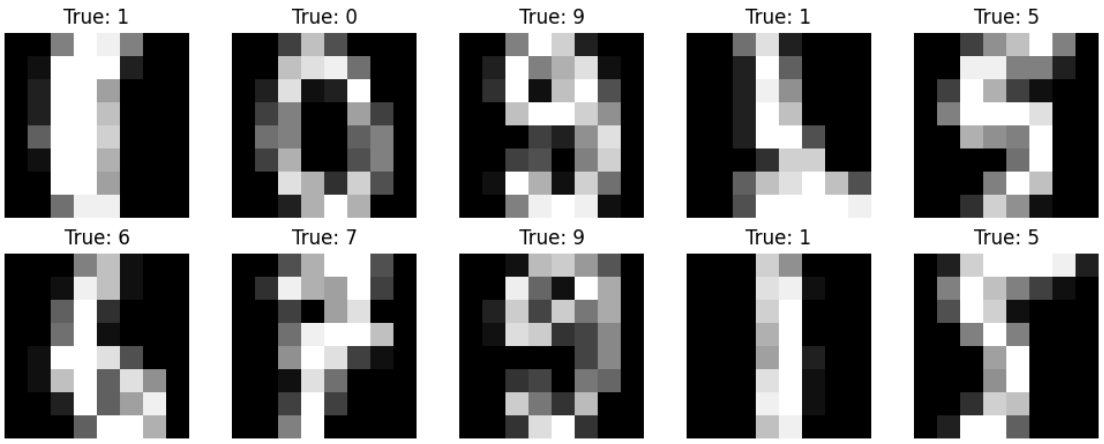
    accuracy                           0.98       450
   macro avg       0.98      0.98      0.98       450
weighted avg       0.98      0.98      0.98       450
```

In [3]:

```
ConfusionMatrixDisplay.from_predictions(y_test,pred,cmap="Blues")
plt.title("SVM - handwritten digits"); plt.show()

# Display first 10 held-out images by original row index
X_train_idx, X_test_idx, y_train2, y_test2 = train_test_split(
    range(len(digits.data)), digits.target, test_size=.25, random_state=42,
    stratify=digits.target)
fig,axes=plt.subplots(2,5,figsize=(10,4))
for ax,idx in zip(axes.ravel(),X_test_idx[:10]):
    ax.imshow(digits.images[idx],cmap="gray")
    ax.set_title(f"True: {digits.target[idx]}"); ax.axis("off")
plt.tight_layout(); plt.show()
```




True: 1
True: 0
True: 9
True: 1
True: 5
True: 6
True: 7
True: 9
True: 1
True: 5